

Transitivity $\wedge$ Independence $\wedge$ Closed-graph continuity $\Rightarrow$ Expected multi-utility representation

Result — human; produced by literature.

0.1 Premises

- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.
- **Independence.** For all $p, q, r \in \Delta(X)$ and $\lambda \in (0,1]$: $p \succcurlyeq q$ if and only if $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$.
- **Closed-graph continuity.** The set $\{(p, q) \in \Delta(X) \times \Delta(X) : p \succcurlyeq q\}$ is closed in the product topology.

0.2 Conclusion

- **Expected multi-utility representation.** There is a set U of functions $u : X \rightarrow \mathbb{R}$ such that for all $p, q \in \Delta(X)$: $p \succcurlyeq q$ if and only if $EU_u(p) \geq EU_u(q)$ for every $u \in U$.

0.3 Proof

Dubra–Maccheroni–Ok, Theorem 1, for compact metric X (finite X is a special case): a reflexive transitive relation on $\Delta(X)$ satisfying independence and closed-graph continuity has an expected multi-utility representation. The proof identifies $\succcurlyeq$ with a closed convex cone of signed measures and takes U to be the dual cone.

0.4 Notes

Reflexivity is a standing assumption of the framework.

0.5 Sources

- Dubra, Maccheroni & Ok (2004), Expected utility theory without the completeness axiom, JET 115, Theorem 1

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